

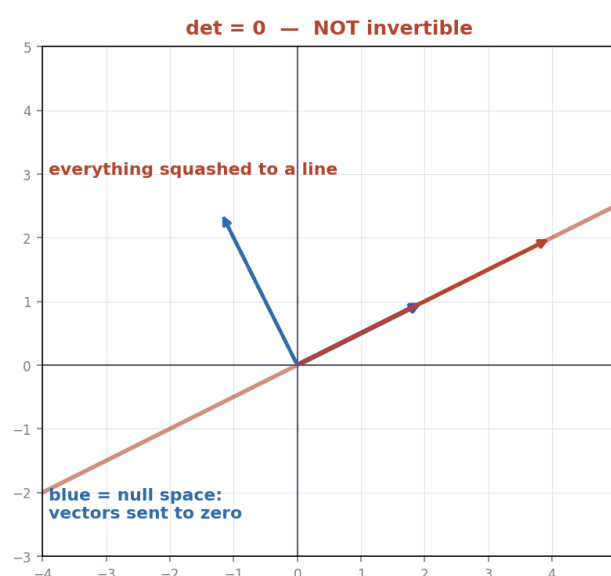
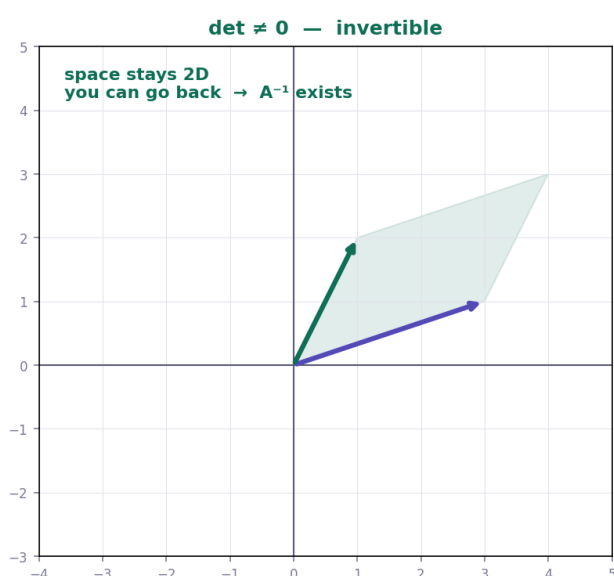
Inverse matrices, column space and null space

Undoing a transformation — and what happens when you can't.

THE IDEA

If a transformation moves space around without collapsing it, there is another transformation that **undoes** it — the **inverse**, written A^{-1} . Applying A then A^{-1} leaves everything exactly where it started. This is how you solve systems of equations: $A\mathbf{x} = \mathbf{b}$ becomes $\mathbf{x} = A^{-1}\mathbf{b}$.

The one condition: an inverse exists only when **det $\neq 0$** . If $\det = 0$, space has been squashed into a lower dimension and information is destroyed — two different inputs land on the same output, so there is no way to reverse it.



THREE WORDS YOU NEED

Term	Meaning
Column space	The span of the matrix's columns — all outputs the matrix can actually produce
Rank	The number of dimensions in that output. Rank 2 in 2D = "full rank"
Null space	All the vectors that get sent to zero . Also called the kernel

When $\det \neq 0$ the null space contains only the origin — nothing is lost. When $\det = 0$ a whole line (or plane) is crushed to zero, and that line *is* the null space.

THE INVERSE FORMULA (2×2)

Swap a and d, flip the signs of b and c, divide everything by the determinant.

WORKED EXAMPLES

CHAPTER 7 — WORKED EXAMPLES

THE INVERSE FORMULA (2×2)

$$\text{for } \begin{bmatrix} a & b \\ c & d \end{bmatrix} : A^{-1} = 1/\det \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

swap a and d, flip the signs of b and c, divide by det

EXAMPLE 1 — find the inverse

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\det = (3)(2) - (1)(1) = 5 \quad \det \neq 0 \rightarrow \text{inverse exists}$$

$$\begin{aligned} A^{-1} &= 1/5 \times \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2/5 & -1/5 \\ -1/5 & 3/5 \end{bmatrix} \end{aligned}$$

EXAMPLE 2 — no inverse

$$B = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

$$\det = (2)(2) - (4)(1) = 0$$

→ **columns dependent, space collapses, NO inverse**

RANK & NULL SPACE

rank = number of dimensions in the output (column space)

null space = all vectors that get sent to zero

full rank → null space is just the origin → invertible

TEST — Chapter 7

1. A has columns (4, 0) and (0, 2). Find det, then A^{-1} . → **det = 8; A^{-1} columns (1/4, 0) and (0, 1/2)**
2. A has columns (3, 1) and (2, 1). Find det, then A^{-1} . → **det = 1; A^{-1} = columns (1, -1) and (-2, 3)**
3. B has columns (6, 3) and (2, 1). Does B^{-1} exist? Justify with a number. → **det = 6(1) - 2(3) = 0 → no inverse; columns dependent**

FORMULAS FROM THIS CHAPTER

Inverse exists	only if $\det \neq 0$
Inverse (2×2)	$A^{-1} = (1/\det) \times [d, -b ; -c, a]$
Solving	$Ax = b \rightarrow x = A^{-1}b$
Rank	dimensions of the output (column space)
Null space	all vectors sent to zero